Class Test 2

Electricity and Electronics EBN 111

11 May 2023

Test Information			
Maximum marks:	16	Full marks:	15
Duration of test:	30 min + (5 min before and 5 after for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		8	

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Assistant Lecturers: S. Twala, G. Neeuwfan, B. Mahlakoana, M. Modiba

ECSA outcome assessment

The following ECSA graduate attributes are assessed at entry level in this question paper: GA1 and GA2

Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to row 01 of the OCR form, etc.
4. Read the instructions on the OCR form carefully.
5. Answers must include units where applicable.
6. Final answers must be written as real numbers and not as fractions.
7. Use at least 4 significant figures to write numbers, including complex numbers.

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the next 35 minutes of the test. (0-35 min)

Do step 2 within the next 5 minutes. (35-40 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

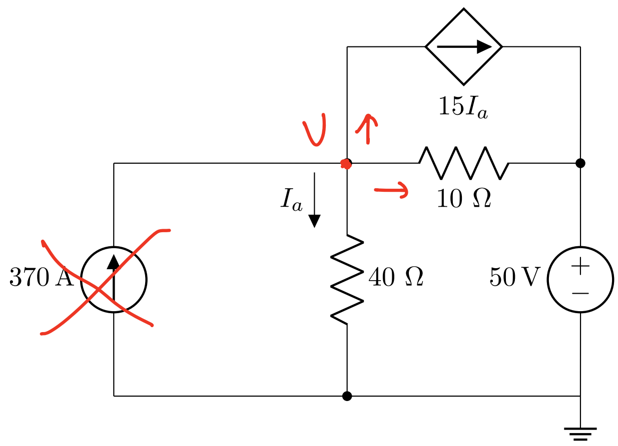
STEP 1 (0 min to 35 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (35 min to 40 min): Carefully copy the answers to the OCR form.

- The Assessment ID is: **2023EBN111C02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Question 1 [2]

For the circuit shown below, calculate the exclusive contribution (denoted by I_a^V) to I_a from the independent voltage source.



01	I_a^V [2]

Switch off the 370A (↑), it becomes an open circuit.

Use nodal analysis for V:

$$I_a + \frac{V-50}{10} + 15I_a = 0, \quad I_a = \frac{V}{40}$$

$$\Rightarrow 16 \cdot \frac{V}{40} + \frac{V-50}{10} = 0$$

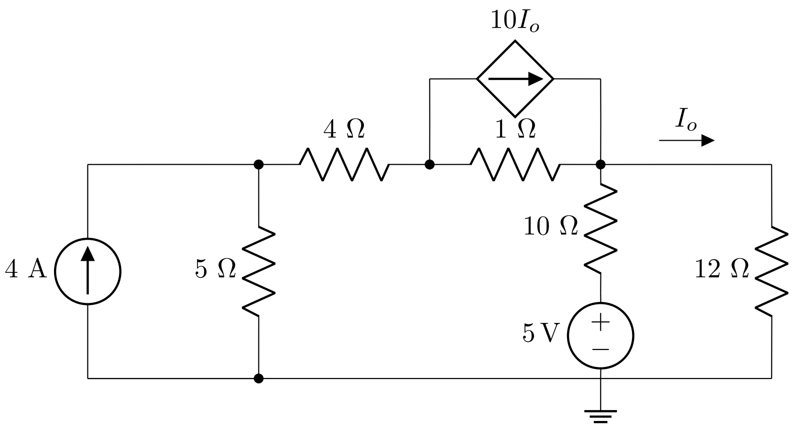
$$16V + 4V - 200 = 0 \Rightarrow V = 10\text{ V}$$

$$I_a^V = \frac{10}{40} = 0.25\text{ A}.$$

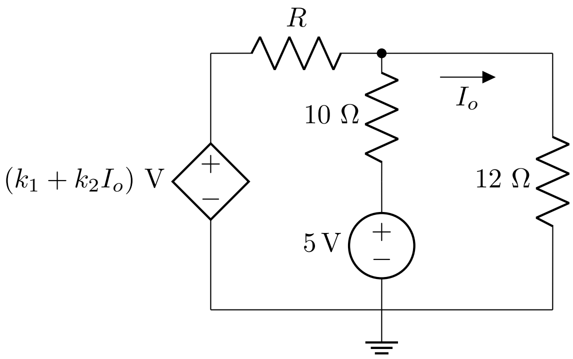
Questions 2 to 3 [2]

Use source transformation to transform Circuit A to Circuit B. Determine k_1 and k_2 in Circuit B.

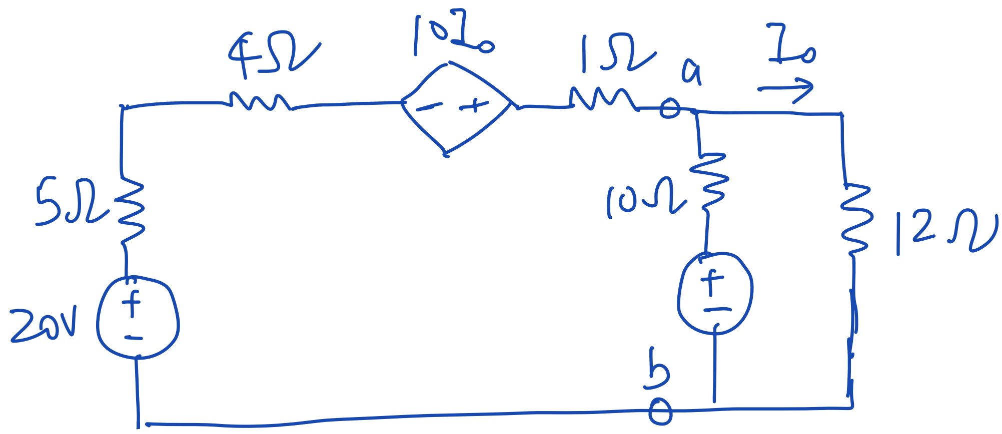
Circuit A:

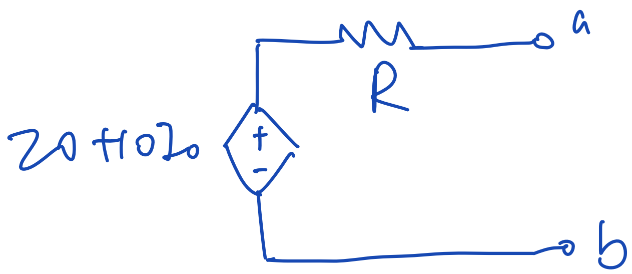


Circuit B:

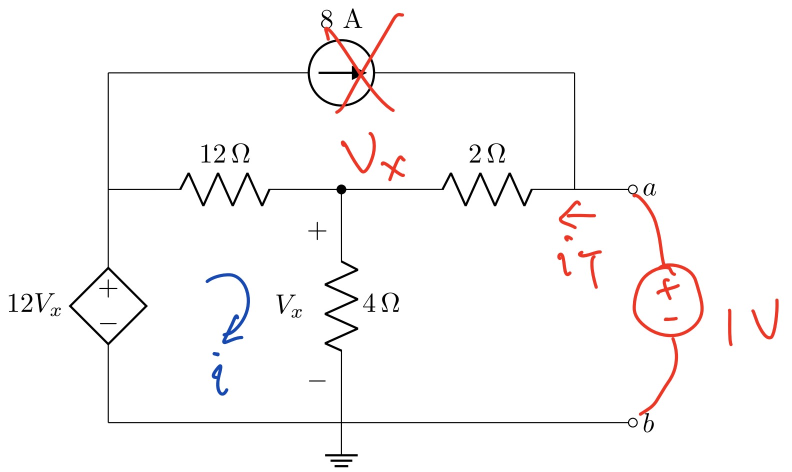


02	k_1 [1]	$k_1 = 20$
03	k_2 [1]	$k_2 = 10$





Questions 4 to 5 [4]	
Find the Thévenin equivalent voltage V_{Th}, and the Thévenin equivalent resistance R_{Th}, with respect to terminals a-b.	



04	V_{Th} [2]
05	R_{Th} [2]

For V_{Th}, use mesh analysis :

$$-12V_x + (i-8)\times 12 + 4i = 0, \quad V_x = 4i$$

$$-48i + 12i - 96 + 4i = 0$$

$$\Rightarrow -32i = 96 \Rightarrow i = -3\text{ A.}$$

$$V_{Th} = 8\times 2 + V_x = 8\times 2 + 4(-3) = 4\text{ V.}$$

For R_{Th}: use nodal analysis:

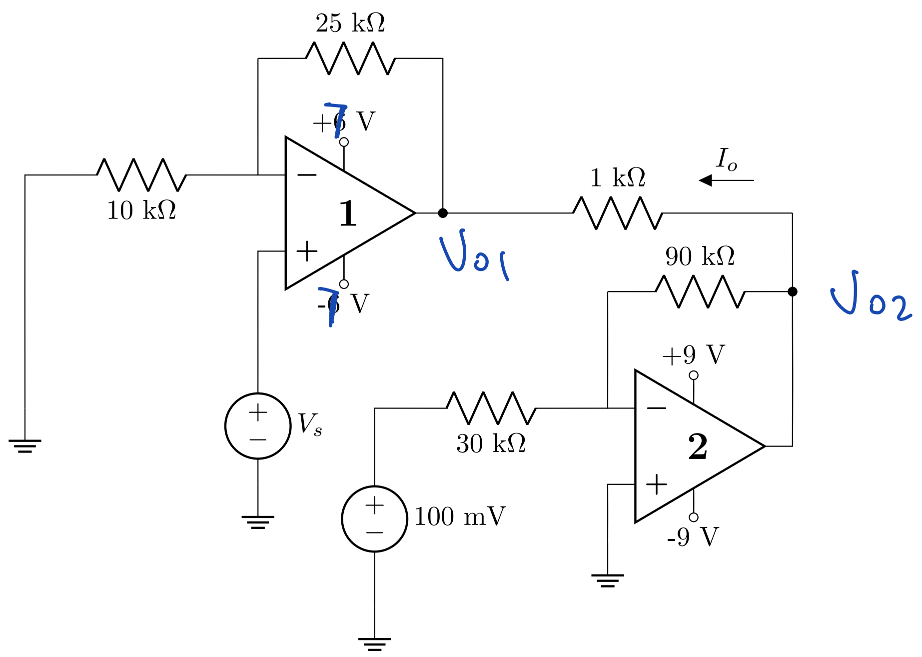
$$\frac{V_x - 12V_x}{12} + \frac{V_x}{4} + \frac{V_x - 1}{2} = 0$$

$$\Rightarrow -11V_x + 3V_x + 6V_x - 6 = 0$$

$$\Rightarrow -2V_x = 6, \quad V_x = -3,$$

$$i_T = \frac{1-V_x}{2} = 2\text{A.}$$

$$R_{Th} = \frac{1}{i_T} = 0.5\Omega$$

Questions 6 to 7 [4]	
	
06	Calculate I_o if $V_s = 1$ V. [2]
07	For what value of V_s will operational amplifier 1 enter the positive saturation? [2]

06: Op Amp 1: non-inverting Op Amp:

$$V_{o1} = V_s \cdot \left(1 + \frac{25k}{10k}\right) = 3.5 \text{ V}$$

OpAmp 2: inverting op Amp:

$$V_{o2} = -0.1 \times \frac{90k}{30k} = -0.3 \text{ V}.$$

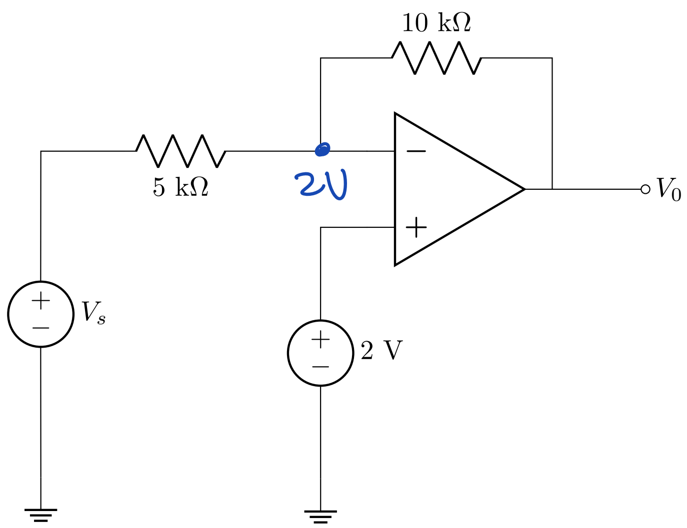
$$I_o = \frac{V_{o2} - V_{o1}}{1k\Omega} = \frac{-0.3 - 3.5}{1k} = -3.8 \text{ mA}.$$

07: when $V_{o1} = V_s\left(1 + \frac{25k}{10k}\right) \geq 7$

$$\Rightarrow V_s \cdot 3.5 \geq 7$$

$$\Rightarrow V_s = 2V.$$

Question 8 [2]



08	Determine V_o when $V_s = 4$ V. [2]

$$\frac{4-2}{5K} = \frac{2-V_0}{10K}$$

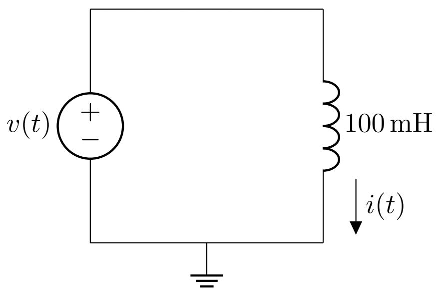
$$\Rightarrow \quad 4 = 2 - V_0 \Rightarrow V_0 = -2 \text{ V.}$$

Question 9 [2]

The voltage signal applied to the 100 mH inductor shown in the circuit below is given by the expression:

$$v(t) = 20e^{-10t} \text{ V}.$$

Find the inductor current at $t = 0.1$ s. Assume that the inductor is not charged initially.



09	$i(0.1 \text{ s})$ [2]

$$i(t) = \frac{1}{L}\int_0^t v(t)\,dt + V(0)$$

$$= \frac{1}{0.1}\int_0^t 20e^{-10t}\,d(-10t) \cdot \frac{1}{-10}$$

$$= -20e^{-10t}\Big|_0^t$$

$$= 20 - 20e^{-10t}$$

$$i(0.1s) = 20 - 20e^{-1} = 12.642A$$

Grand Total : [16]

PRACTICE ANSWER SHEET

Assessment ID	2	0	2	3	E	B	N	1	1	1	C	0	2

Please round all irrational numbers to a minimum of 5 significant digits.

	Answer	Unit
01	$I_a^V =$ 0.25	A
02	$k_1 =$ 20	No unit required
03	$k_2 =$ 10	No unit required
04	$V_{Th} =$ 4	V
05	$R_{Th} =$ 0.5	Ω
06	$I_o =$ −3.8	mA
07	$V_s =$ 2	V
08	$V_o =$ −2	V
09	$i(0.1\text{ s})=$ 12.642	A